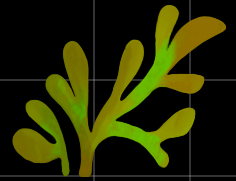
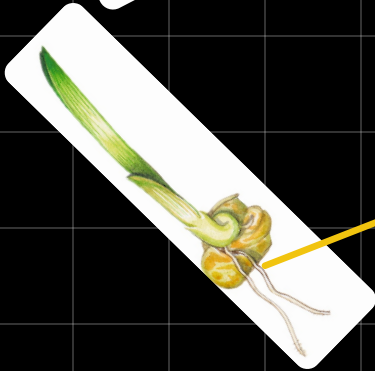




PLANT MORPHOLOGY



- Direct elongation of radicle leads to formation of primary root. It bears secondary, tertiary roots etc.
- The primary roots and its branches form **Tap root system**.
- e.g. Mustard, Mango, Pea.

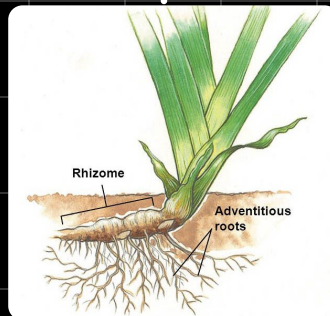


- Primary root is short lived and is replaced by large number of roots.
- These roots originate from the base of the stem and constitute the **fibrous root system**.
- e.g. Wheat, Rice, Maize.

ADVENTITIOUS ROOT



Banyan

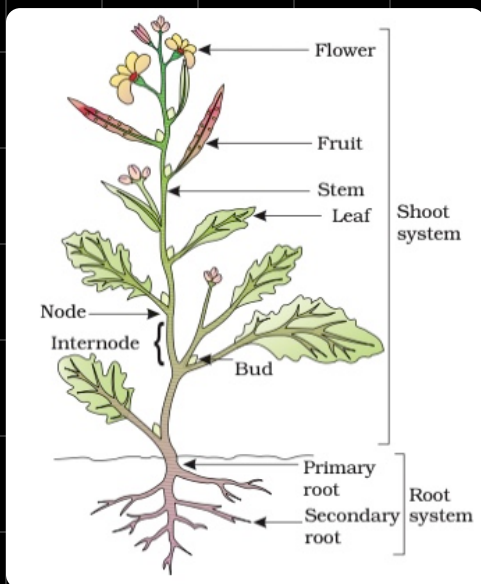


Grass



Monstera

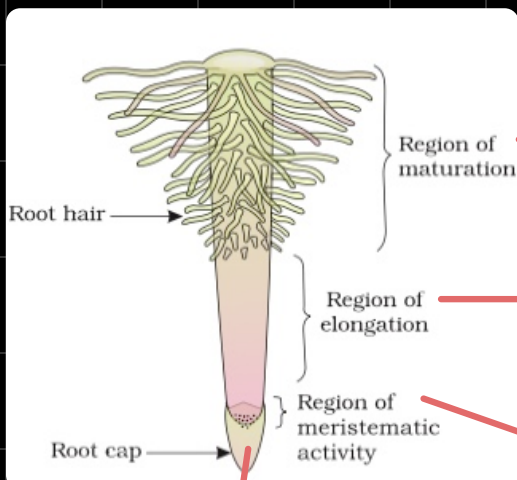
Roots arise from parts other than radicle



Functions of root:

1. Water and minerals absorption.
2. Providing anchorage.
3. Storing reserve food materials.
4. Synthesis of plant growth regulators.

Parts of root:



- Present at apex.
- Thimble like structure.
- Protects the apical portion of root.

- Root hairs formed by some epidermal cells.
- Root hairs absorb water and minerals from soil.

- Present proximal to meristematic region.
- Rapid elongation and enlargement.
- Responsible for length growth.

- Present above root cap.
- Small, thin walled cells, dense protoplasm.
- Repeated division

Root modifications

Storage

Tap root



Carrot



Turnip

Adventitious root



Sweet potato

Support

Prop roots



Banyan tree

Stilt roots



Sugarcane,
Maize

Breathing roots



Rhizophora

STEM

1. It bears branch, leaf, flower, fruit.
2. Develops from plumule.
3. Bear nodes and internodes.



4. Bear terminal and axillary buds.
5. Conduct water, minerals, photosynthates.
6. Perform storage, support, protection and vegetative propagation.

Stem modifications

Underground

Storage



Potato



Colocasia



Ginger

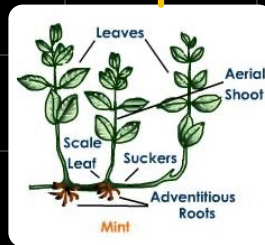


Turmeric



Zaminkand

Sub aerial



In Mint and Jasmine, lateral branch arising from base of main axis arch downward and touch the ground.

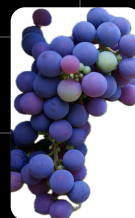
Aerial

Support

Tendrils



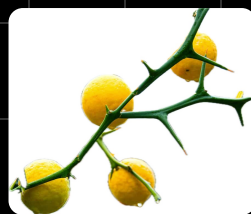
Gourds



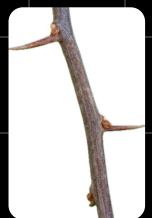
Grapevine

Protection

Thorns



Citrus

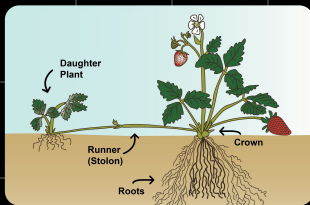


Bougainvillea

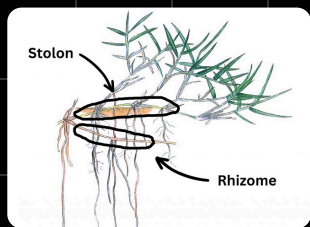
Stem modifications

Underground

Propagation

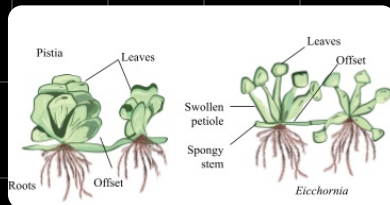


Strawberry



Grass

Sub aerial



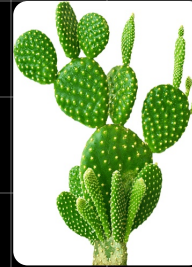
Pistia

Eicchornia

Lateral branch with short internodes and each node bearing a rosette of leaves and a tuft of roots.

Aerial

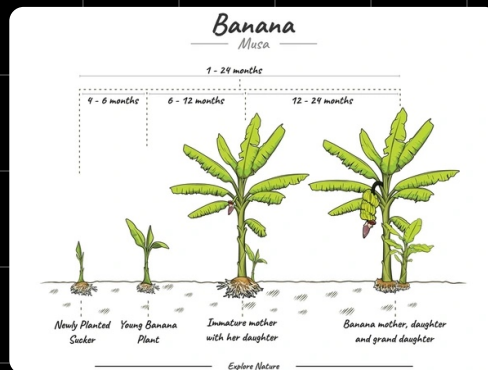
Photosynthetic



Opuntia



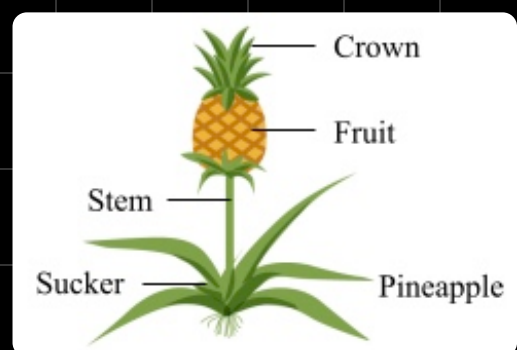
Euphorbia



In banana, pineapple and Chrysanthemum, the lateral branches originate from the basal and underground portion of the main stem, grow horizontally beneath the soil and then come out obliquely upward giving rise to leafy shoots.



Chrysanthemum



Pineapple

LEAF

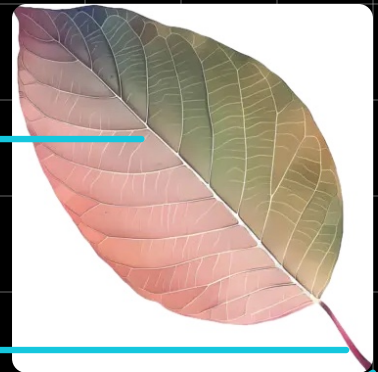


Leaf is lateral. flat structure born on stem. It grows at node and bear bud in its axil. The axillary bud develops into branch. They develop from SAM, and arranged in acropetal order. They perform photosynthesis.

Parts of leaf

Lamina or the leaf blade is the green expanded part of the leaf with veins and veinlets. Veins provide rigidity to the leaf blade and act as channels of transport for water, minerals

Lamina



Petiole

Leaf base

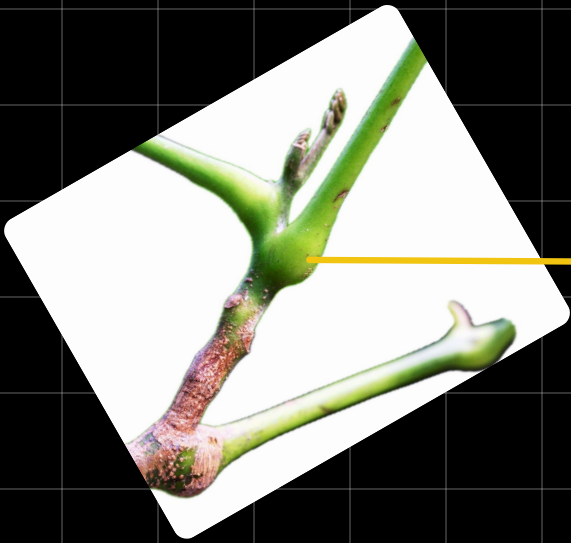
The petiole help hold the blade to light. Long thin flexible petioles allow leaf blades to flutter in wind, thereby cooling the leaf and bringing fresh air to leaf surface.

leaf is attached to the stem by the leaf base, and may have two lateral stipules.



Sheathing
Leaf base

In monocotyledons, the leaf base expands into a sheath covering the stem partially or wholly.



In some leguminous plants the leafbase may become swollen, known as **pulvinus**.

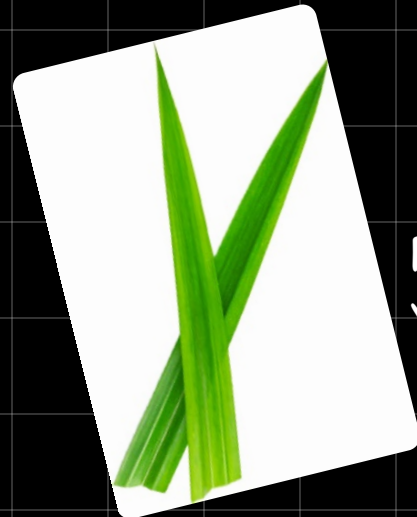
Venation

The arrangement of veins and veinlets.



Veinlets form the network.
e.g. Dicots

Reticulate venation



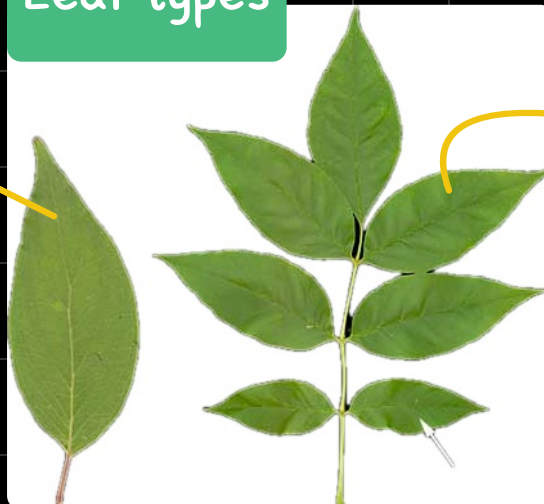
Parallel venation

Veins run parallel to each other within a lamina.

Leaf types

Lamina is entire or when incised, the incisions do not touch the midrib.

Simple leaf



Incisions of the lamina reach up to the midrib breaking it into a number of leaflets.

Bud is only present in axil of petiole and not leaflet.

Compound leaf

Compound Leaf types

Number of leaflets are present on a common axis, the rachis, which represents the midrib of the leaf. e.g. Neem.



Pinnately compound



The leaflets are attached at a common point, i.e., at the tip of petiole. e.g. Silk cotton.

Palmately compound

Pattern of arrangement of leaves on the stem or branch.

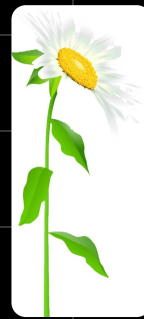
Phyllotaxy



China rose



Mustard



Sunflower

ALTERNATE



Calotropis

OPPOSITE



Guava

WHORLED



Alstonia

Modifications of Leaves

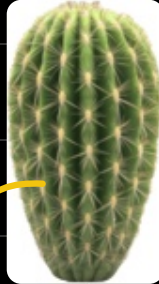
CLIMBERS



Tendrils

Tendrils for climbing.
e.g. peas

SPINES



Spines for defense.
e.g. Cactus

BULB



Fleshy leaves.
e.g. Onion, Garlic

INSECTIVOROUS



Leaves modified to trap insects and manage Nitrogen deficiency.
e.g. Pitcher plant, Venus fly trap.



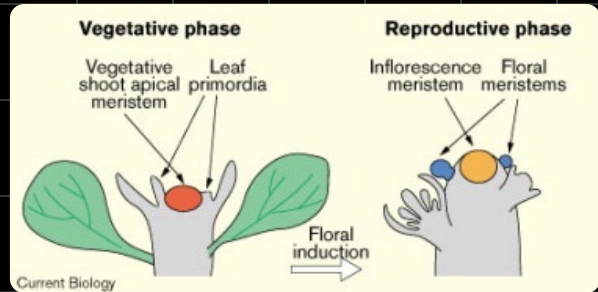
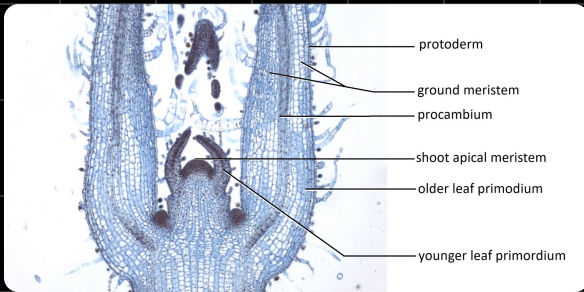
PHYLLODE



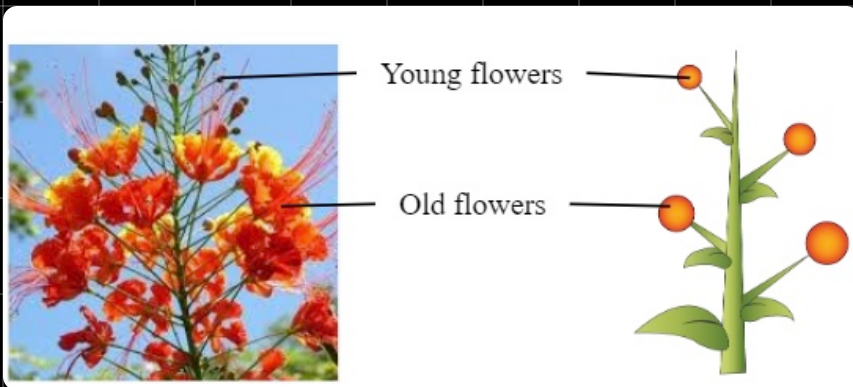
Petiole

Leaves are small and short-lived.
The petioles in these plants expand, become green and synthesise food.
e.g. Australian acacia

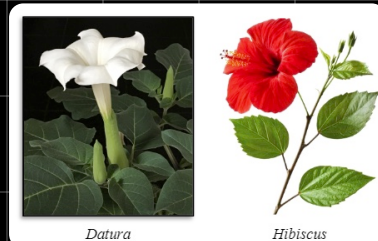
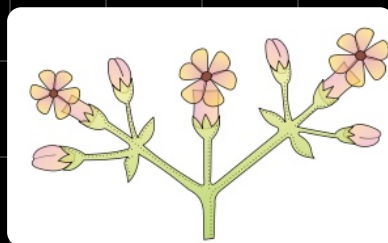
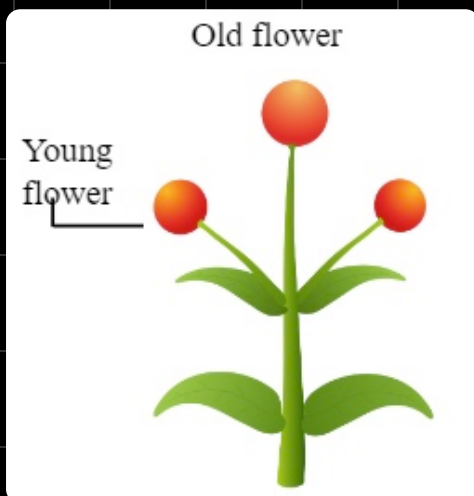
INFLORESCENCE



- Shoot Apical Meristem changes to floral meristem.
- Internodes do not elongate and the axis gets condensed.
- The apex produces different kinds of floral appendages laterally at successive nodes instead of leaves.
- The arrangement of flowers on the floral axis is called inflorescence.



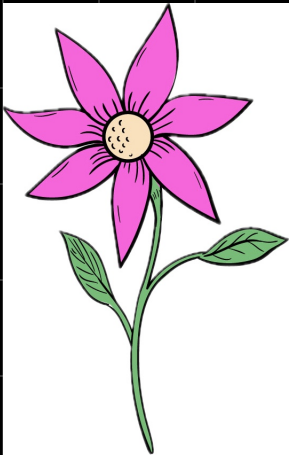
In racemose type of inflorescences the main axis continues to grow, the flowers are borne laterally in an acropetal succession.



In cymose type of inflorescences the main axis terminates in a flower, hence is limited in growth.

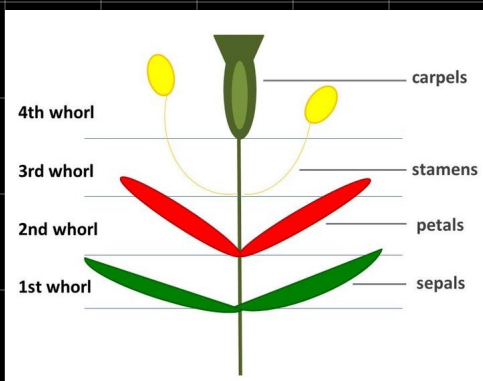
The flowers are borne in a basipetal order

FLOWER



- It is the reproductive unit in angiosperm.
- It is meant for sexual reproduction.
- The flower has four different kinds of whorls arranged on the swollen end of **pedicel** called **receptacle/thalamus**.

FLOWER WHORLS



Androecium and **Gynoecium** are reproductive organs.

Calyx and **Corolla** are accessory organs.



Perianth

Calyx and Corolla not distinct.
e.g. Lily

BISEXUAL FLOWER

UNISEXUAL FLOWER

Pistillate
flower



Staminate
flower



Carpel
Stamen

FLOWER SYMMETRY

Actinomorphic



Mustard



Datura



Chilli

Flower can be divided into two equal halves in any radial plane passing through centre.
aka Radial symmetry.

Zygomorphic



Pea



Bean

Flower can be divided into two equal halves only in one vertical plane.
aka Bilateral symmetry.
e.g. Pea, Gulmohar, Bean, Cassia.

Asymmetric



Canna

Flower cannot be divided into two equal halves in any vertical plane.



Bract

Reduced leaf found at the base of the pedicel.



Trimerous

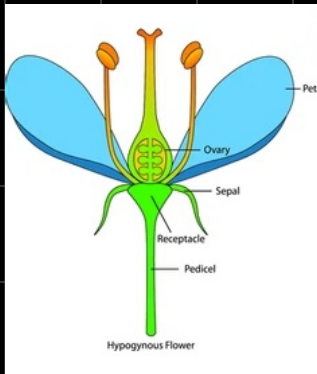


Tetramerous

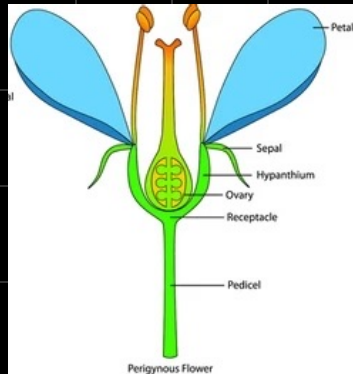


Pentamerous

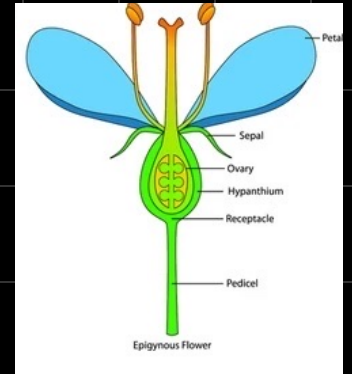
FLOWER POSITION ON THALAMUS



Gynoecium occupy highest position. other whorls arise below it.
e.g. Mustard, China rose, Brinjal.



Gynoecium situated in the centre. Other whorls located on the rim of thalamus.
e.g. Plum, Rose, Peach.



Thalamus margin grows upward enclosing the ovary completely and fusing with it. Other parts arise above the ovary.
e.g. Guava, Cucumber, ray florets Sunflower.

FLOWER PARTS



COROLLA

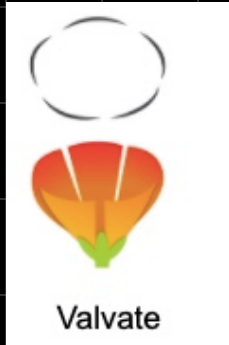
Member called petals. Brightly colored to attract insects.
Gamopetalous (fused petals)
Polypetalous (petals free)

CALYX

Outermost whorl of a flower, members called sepals. Protect flower in bud stage.
Gamospealous (fused sepals)
Polysepalous (sepals free)

AESTIVATION

Mode of arrangement of sepals or petals in floral bud with respect to the other members of the same whorl is known as aestivation.



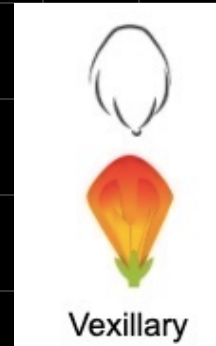
sepals or petals in a whorl just touch one another at the margin, without overlapping.
e.g. Calotropis



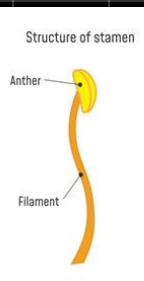
one margin of the appendage overlaps that of the next one and so on.
e.g. China rose, Cotton, Ladyfinger



margins of sepals or petals overlap one another but not in any particular direction.
e.g. Cassia, Gulmohar

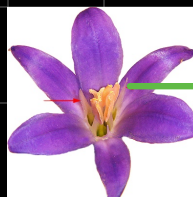


There are five petals, the largest (standard) overlaps the two lateral petals (wings) which in turn overlap the two smallest anterior petals (keel).
e.g. Pea, Bean



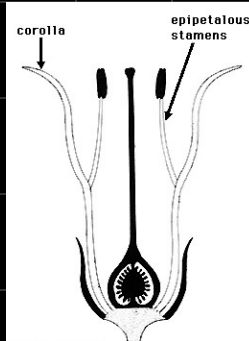
Androecium consist of stamens.

Stamen made up of stalk called filament and anther. Anther is bilobed, tetrasporangiate with four pollen sacs. Pollen grains produced in pollen sac.



Sterile stamen (staminode)

Epipetalous

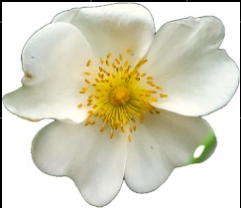


stamens are attached to the petals, they are **epipetalous**.
e.g. brinjal

Epiphyllous



Stamens attached to the perianth, they are **epiphyllous**.
e.g. lily



stamens in a flower may remain free (polyandrous).
e.g. Rose, Lotus



Monadelphous



stamens may be united into one bundle.
e.g, China rose

Diadelphous



stamens may be united into two bundles.
e.g, Pea

Tetradynamous



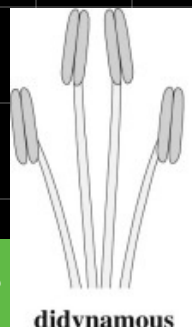
Two outer short stamen,
Four inner long stamen.
e.g. Mustard

Polyadelphous

stamens may be united into more than two bundles.
e.g, Citrus



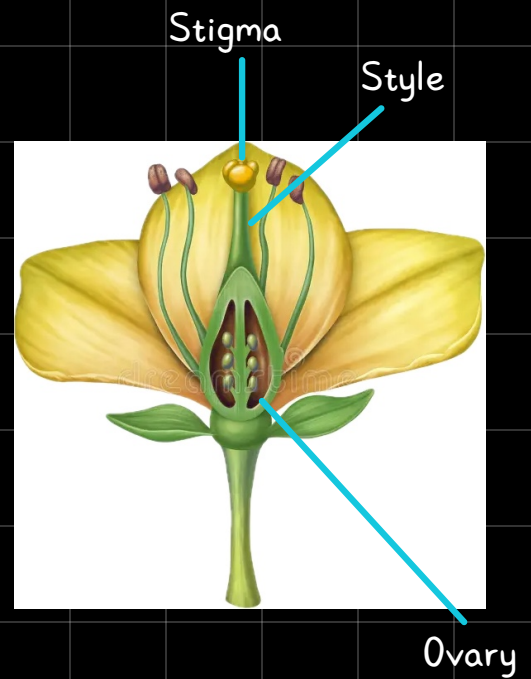
Didynamous



four stamens in two pairs of unequal length.
e.g. Salvia

GYNOECIUM

- Female reproductive part of the flower and is made up of one or more carpels.
- Carpel consists of three parts namely stigma, style and ovary.
- **Ovary** is the enlarged basal part, on which lies the elongated tube, the style.
- The **style** connects the ovary to the stigma.
- The **stigma** is usually at the tip of the style and is the receptive surface for pollen grains.



Locule

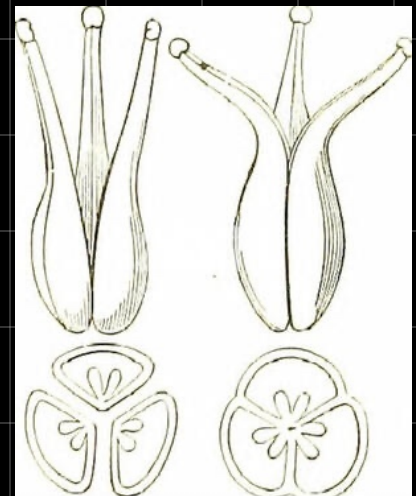
Ovule

Placenta

ovary bears one or more ovules attached to a flattened, cushion-like placenta.

More than one carpel is present, they may be free (as in lotus and rose) and are called **apocarpous**.

They are termed **syncarpous** when carpels are fused, as in mustard and tomato.



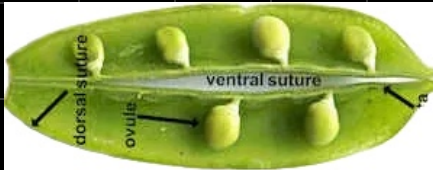
apocarpous

syncarpous

PLACENTATION

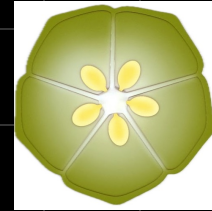
Arrangement of ovules within ovary.

MARGINAL



Placenta forms a ridge along the ventral suture of the ovary and the ovules are borne on this ridge forming two rows.
e.g. Pea

AXILE



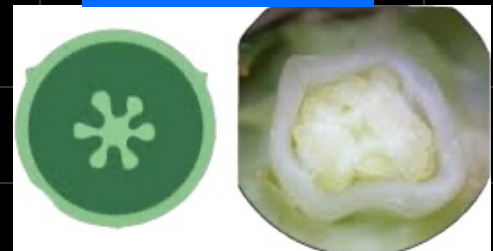
Placenta is axial and the ovules are attached to it in a multilocular ovary.
e.g. China rose, Tomato, Lemon.

PARIETAL



Ovules develop on the inner wall of the ovary or on peripheral part. Ovary is one-chambered but it becomes two chambered due to the formation of the false septum,
e.g. Mustard and Argemone.

FREE CENTRAL



Ovules are borne on central axis and septa are absent,
e.g. Dianthus and Primrose

Placenta develops at the base of ovary and a single ovule is attached to it.
e.g. Sunflower, Marigold.



BASAL

FRUIT

Characteristic feature of the flowering plants. It is a mature or ripened ovary, developed after fertilisation.

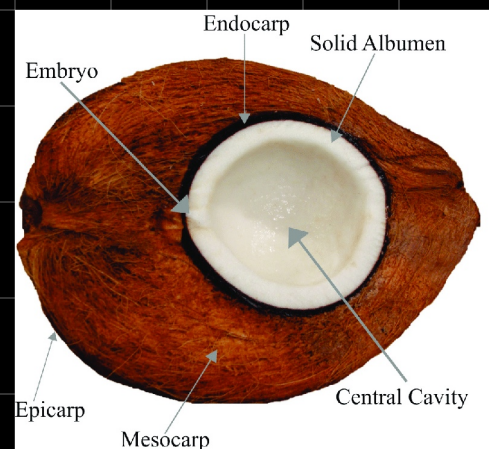
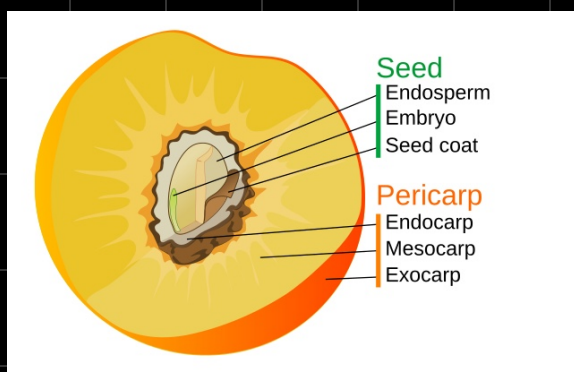
If a fruit is formed without fertilisation of the ovary, it is called parthenocarpic fruit.



- Fruit consists of a wall or pericarp and seeds.
- The pericarp may be dry or fleshy.
- When pericarp is thick and fleshy, it is differentiated into the outer epicarp, the middle mesocarp and the inner endocarp.

- Mango and coconut, the fruit is known as a drupe.
- It develop from monocarpellary superior ovaries and are one seeded.
- Mango, the pericarp is well differentiated into an outer thin epicarp, a middle fleshy edible mesocarp and an inner stony hard endocarp.

In coconut which is also a drupe, the mesocarp is fibrous.



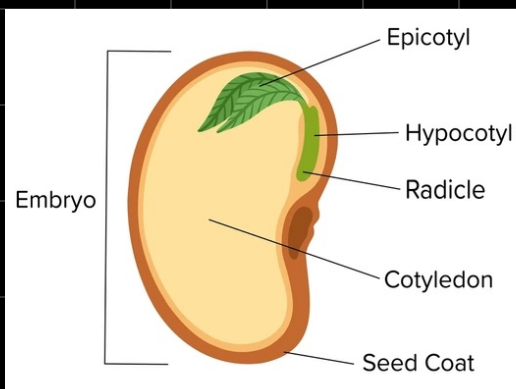
SEED

- Ovules after fertilisation, develop into seeds.
- A seed is made up of a seed coat and an embryo.
- The embryo is made up of a radicle, an embryonal axis and one (as in wheat, maize) or two cotyledons (as in gram and pea).



DICOT SEED

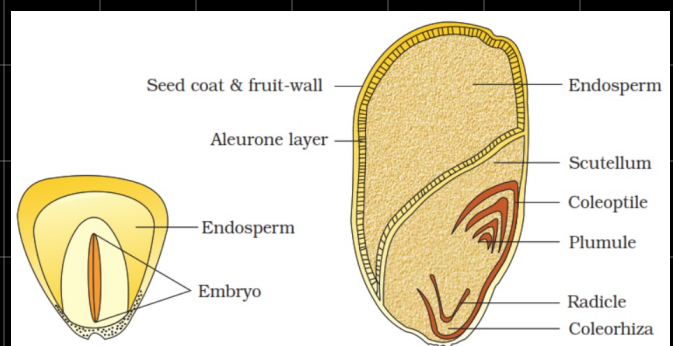
1. Outermost covering of a seed is the **seed coat**.
2. The seed coat has two layers, the outer **testa** and the inner **tegmen**.
3. The **hilum** is a scar on the seed coat through which the developing seeds were attached to the fruit.
4. Above the hilum is a small pore called the **micropyle**.
5. Within the seed coat is the embryo, consisting of an embryonal axis and two cotyledons.
6. The **cotyledons** are often fleshy and full of reserve food materials.
7. At the two ends of the embryonal axis are present the **radicle** and the **plumule**.
8. In some seeds such as **Castor** the endosperm formed as a result of double fertilisation, is a food storing tissue.
9. In plants such as bean, gram and pea, the endosperm is not present in mature seeds and such seeds are called non-endospermous.



Dicot seed

MONOCOT SEED

1. Consists of one large and shield shaped cotyledon known as **scutellum** and a short axis with a plumule and a radicle.
2. The plumule and radicle are enclosed in sheaths which are called **coleoptile** and **coleorhiza**.
3. Seeds are endospermic but some as in orchids are non-endospermic.
4. Seed coat is membranous and generally fused with the fruit wall.
5. The **endosperm** is bulky and stores food.
6. The outer covering of endosperm separates the embryo by a proteinous layer called **aleurone layer**.
7. Embryo is small and situated in a groove at one end of the endosperm.



Monocot seed

FLORAL FAMILY DESCRIPTION

1. Fabaceae

Vegetative characters

1. This family was earlier called Papilionoideae, a subfamily of family Leguminosae.
2. Root with root nodules.
3. Leaves alternate, leaf base pulvinate, stipulate.



Floral characters

1. Racemose.
2. Bisexual, zygomorphic.
3. Pentamerous, gamospealous, valvate/imbricate.
4. Pentamerous, Polypetalous, vexillary.
5. Androecium ten, diadelphous (9)+1.
6. Ovary superior, monocarpellary.
7. Legume, non-endospermic.



Standard

Wings

Keel



Diadelphous



Marginal
placentation

% ♂ K₍₅₎ C₁₊₂₊₍₂₎ A₍₉₎₊₁ G₁

Economic importance

1. Pulses: Gram, arhar, sem, moong, soyabean
2. Edible oil: Soyabean, groundnut
3. Dye: Indigofera
4. Fibres: Sunhemp
5. Fodder: Sesbania, Trifolium
6. Ornamental: Lupin, sweet pea
7. Medicine: Muliathi

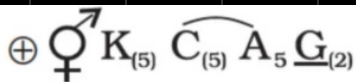
2. Solanaceae

Vegetative characters

1. Herbaceous, solid/hollow, hairy stem
2. Alternate, **exstipulate**, simple leaf

Floral characters

1. Solitary, axillary or cymose
2. Bisexual, actinomorphic
3. Penta, gamospealous, **persistent calyx**,
4. Penta, gamopetalous, valvate
5. Stamens five, **epipetalous**
6. Bicarpellary, syncarpous, hypogynous, bilocular, **swollen placenta**
7. Berry or capsule fruit
8. Endospermic seed



Persistent calyx



Epipetalous



L S Flower



Axile placentation

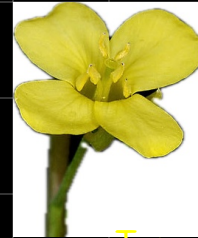
Economic importance

1. Food: Tomato, brinjal, potato
2. Spice: Chilli
3. Medicine: Belladonna, ashwagandha
4. Fumigatory: Tobacco
5. Ornamentals: Petunia

3. Brassicaceae

Vegetative characters

1. Stem herbaceous, solid, branched
2. Leaves simple, alternate, hairy



Tetramerous,
Tetradynamous

Floral characters

1. Racemose
2. Ebracteate, pedicellate, bisexual, actinomorphic, tetramerous, exstipulate
3. polysepalous, 2+2, imbricate
4. Petals 4, polypetalous
5. Stamens 6, polyandrous, tetradynamous 2+4
6. Bicarpellary, syncarpous, bilobed
7. Siliqua fruit
8. Non endospermic seed



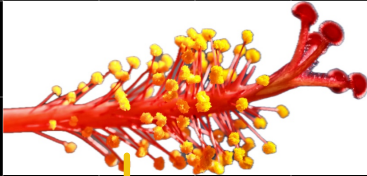
Parietal
placentation

Ebr. $\oplus \begin{matrix} \uparrow \\ \text{♂} \\ \downarrow \\ \text{♀} \end{matrix} K_{2+2} C_4 A_{2+4} G_{(2)}$

Economic Importance

4. Malvaceae

Axillary inflorescence, alternate leaves, bracteate, **epicalyx**, actinomorphic, **axile** placentation, **monoadelphous**, hypogynous, gamosepalous, polypetalous, **epipetalous**

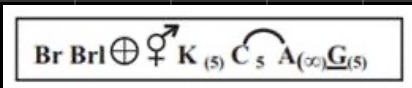


Monoadelphous refers to a condition in flowering plants where the filaments of all the stamens are fused together to form a single bundle.



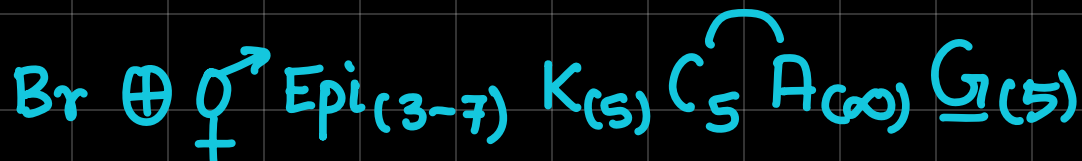
Epicalyx is an additional whorl around the calyx of a flower. It is a group of bracteoles forming the outer whorl of the calyx.

Fruit: Capsule.



Economic Importance

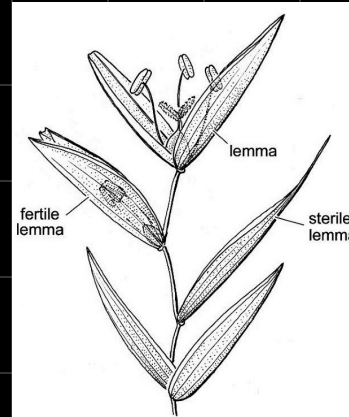
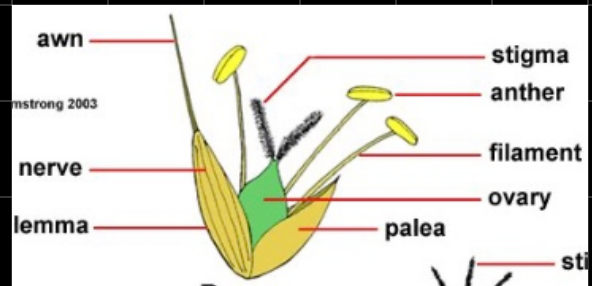
1. Food: Okra, Durian
2. Fibres: Cotton, Jute
3. Beverage: Kola nut
4. Timber: Balsa, Triplochiton
5. Ornamental: Hibiscus
6. Medicinal: Marshmallow root



5. Poaceae

- **Fibrous** root, adventitious root, stilt root (maize), hollow stem, exstipulate, alternate leaves, **parallel venation**, sheathy leaf base, Inflorescence spike/spikelet,
- Male spikelet: tassel.
- Female spikelet: Cob/ear
- Bisexual (rice), Unisexual (maize)
- Perianth with 2 **lodicules**, **polyandrous**, **basal placentation**, **trimerous**, monocarpellary superior ovary.

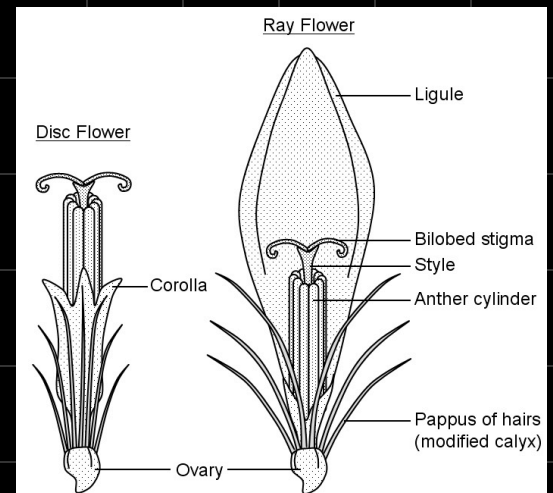
$\% \text{ } \text{♀} \text{ } P_2 \text{ } A_{3+3} \text{ } \underline{G}_1$



6. Asteraceae



Alternate leaves, petiolate, exstipulate, opposite (Helianthus), whorled (Eupatorium), **Capitulum** inflorescence, Unisexual or bisexual, pentamerous, bracteate, epigynous tubular (actinomorphic) or **ligulate**(%)



Ray florets Br. ♂ or neuter K pappus $C(5) A_0 \overline{G(2)}$ or 0.

1. Bracteate, **Zygomorphic**, Unisexual, Sessile,
2. Calyx: pappus like structure
3. Corolla: Petals 5, Gamopetalous,
4. aestivation- 3 or 4 valvate
5. Androecium: **Absent**
6. Gynoecium: Absent or bicarpellary, syncarpous, inferior, unilocular with basal placentation, one anatropous ovule.

Disc florets Br $\text{♂} \text{ } \text{♀}$ K pappus $C_{(5)} A_{(5)} \overline{G_{(2)}}$

1. Bracteate, **Actinomorphic**, bisexual, Inferior ovary
2. Calyx: pappus like structure
3. Corolla: 5 Gamopetalous, valvate aestivation
4. Androecium: 5 **epipetalous**, **Syngenesious stamens**.
5. Gynoecium: Bicarpellary, syncarpous, unilocular with single anatropous ovule, basal placentation, inferior ovary.